



Chapter 2
Network Standards

Also known as protocols



中央資管
陳奕明

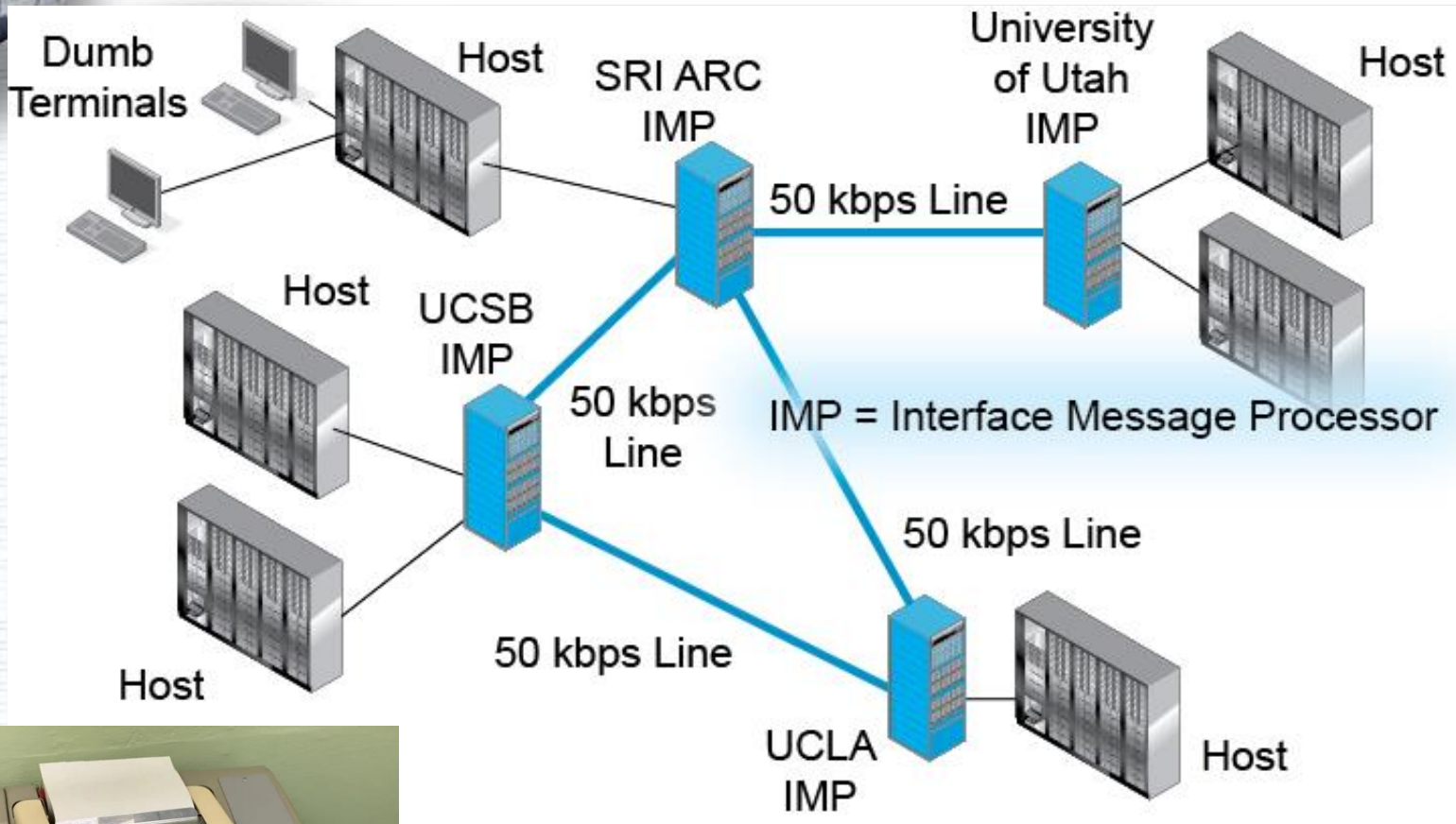
讀完本章，同學將能複習有關網路的基本名詞與重要概念，最後也將補充IoT最新知識

本章的學習需配合Wireshark的練習才能確實瞭解

Network Standards

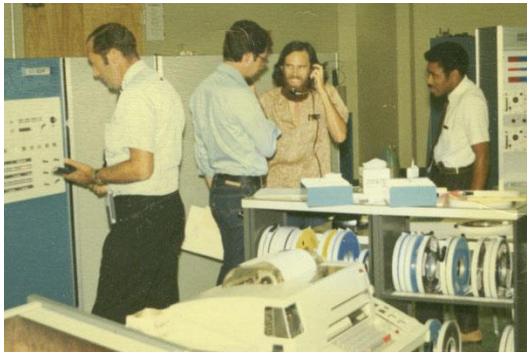
- How the Internet Came to Be
- Introduction
- Message Ordering
- Message Syntax
- Encoding Application Messages
- Vertical Communication on Hosts

2.1 The Early ARPANET -1969正式運行





TELETYPE



The Network Working Group (1/2)

1. Bolt, Beranek, and Newman designed **the IMPs (Interface Message Processor)**
2. But there were **no standards for application programs or protocols for hosts** to communicate with other hosts
3. An **informal group** formed to create needed standards: the Network Working Group(NWG)
4. Mostly **engineers and graduate students**, the NWG had no formal standing

See the following article for more information:

- "**An early history of the internet : History of Communications"**, by **Kleinrock, L.**; in IEEE Communications Magazine, August 2010 Vol. 48, Issue:8 , pp. 26 - 36
- https://www.fastcompany.com/90423457/50-years-ago-today-the-internet-was-born-in-room-3420

The Network Working Group (2/2)

- Called their “standards” **Requests For Comments (RFCs**) instead of standards
- **Informal** and **egalitarian**, working on **rough consensus** rather than voting
- Designed simple standards based on **running code**; **quick and inexpensive to implement**
- The **Internet Engineering Task Force (IETF)** emerged from the NWG, kept many of their characteristics



I E T F®

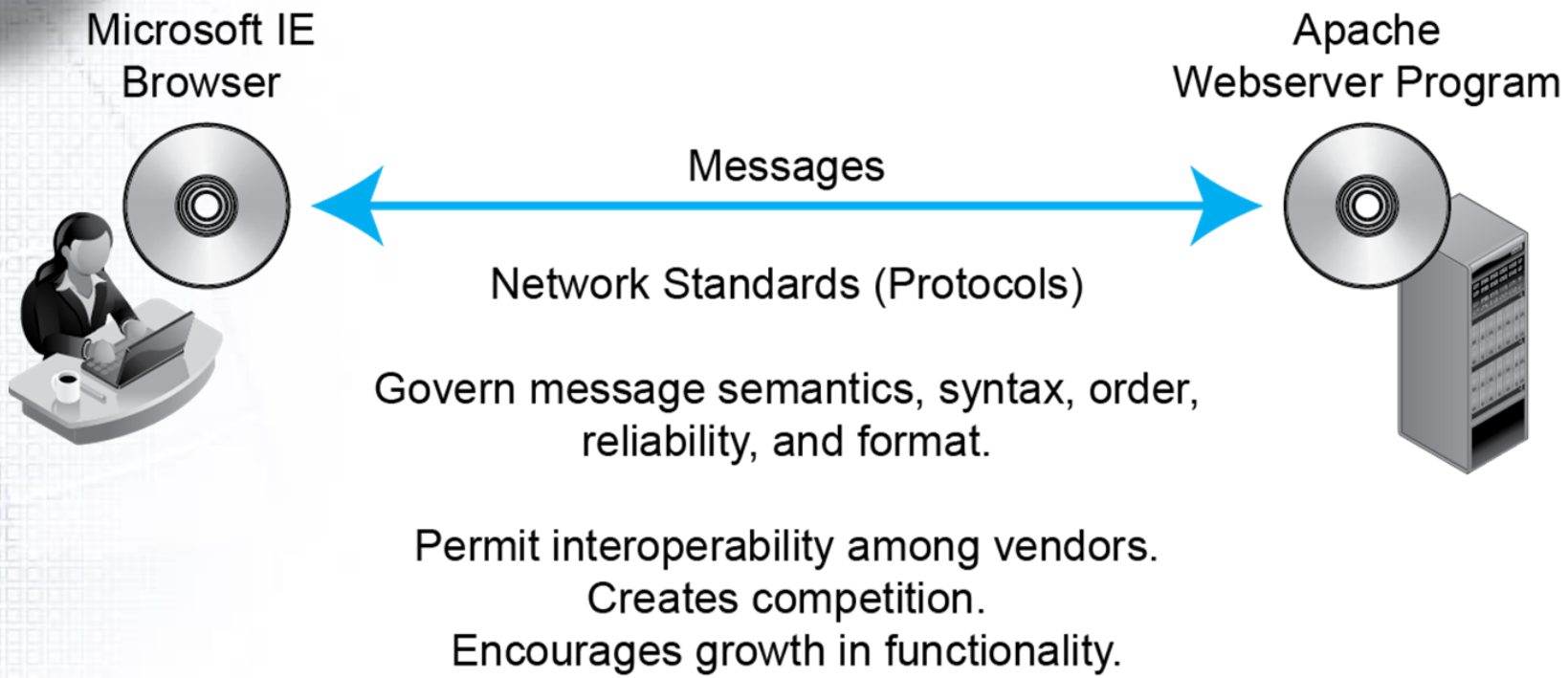
https://www.ietf.org/

Network Standards

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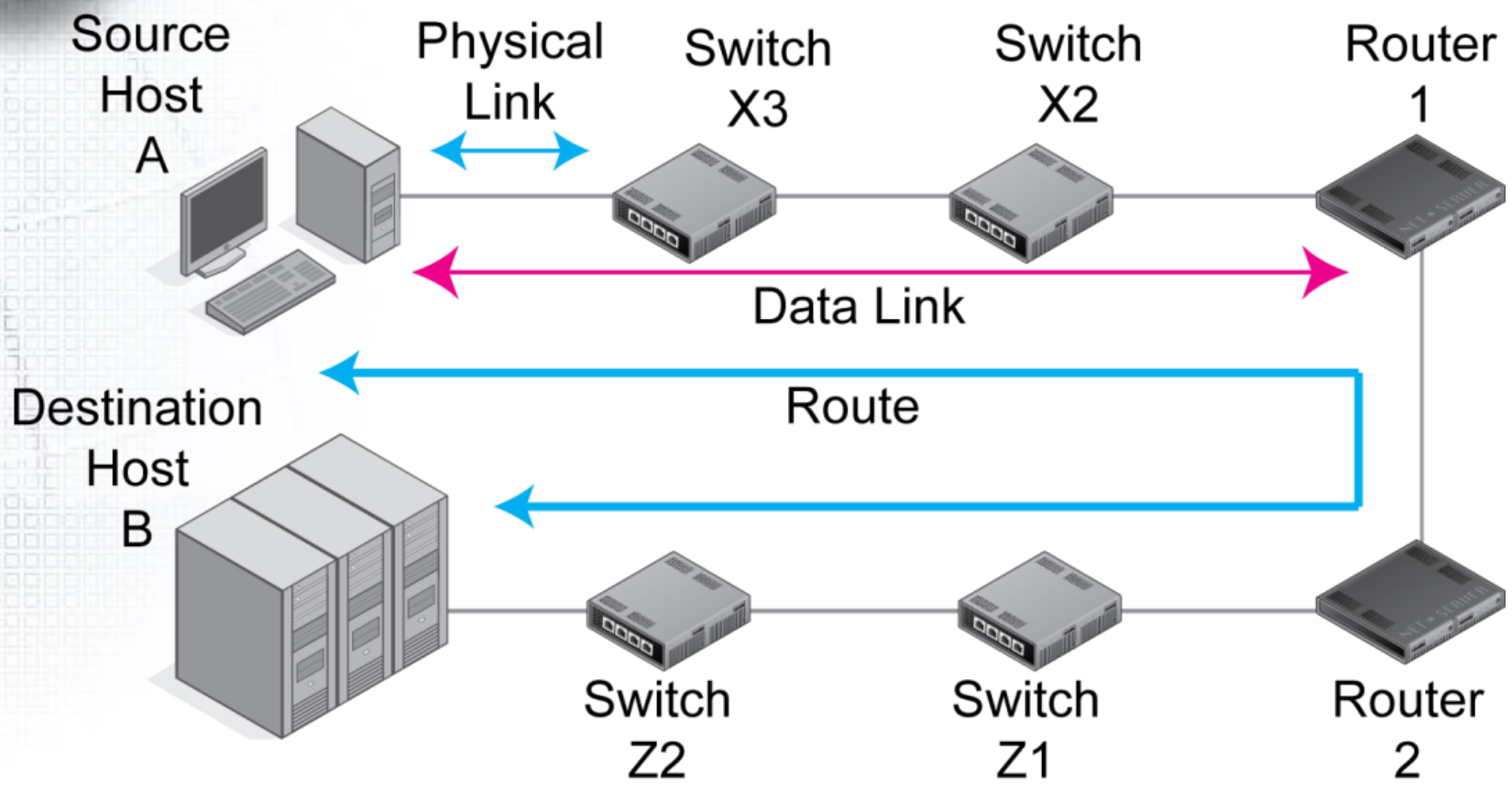


Network Standards-What?



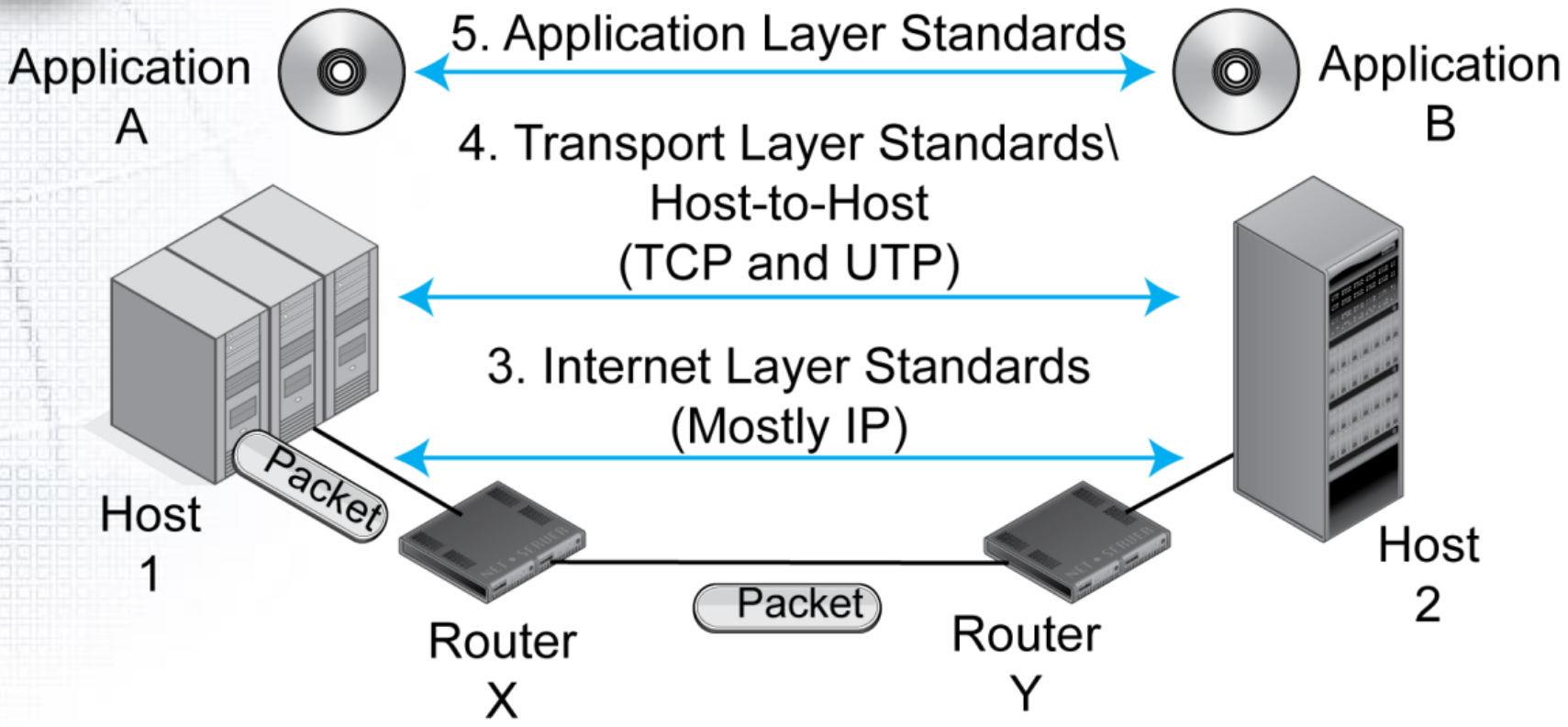
- Why?
  - Bring competition
  - Key to networking--> learn network = learn standards

2.3 Physical, Data Link, and Internet Layers



Q:你能從上圖分辨’physical link’, ‘data link’ 以及’route’三者的區別嗎？

2.4 Transport and Application Processes



Note: In fact, **Transport Layer** provide **Process-to-Process** communication. (i.e. A process's port to B process's port)

Layered TCP/IP Protocol Architecture

TCP/IP

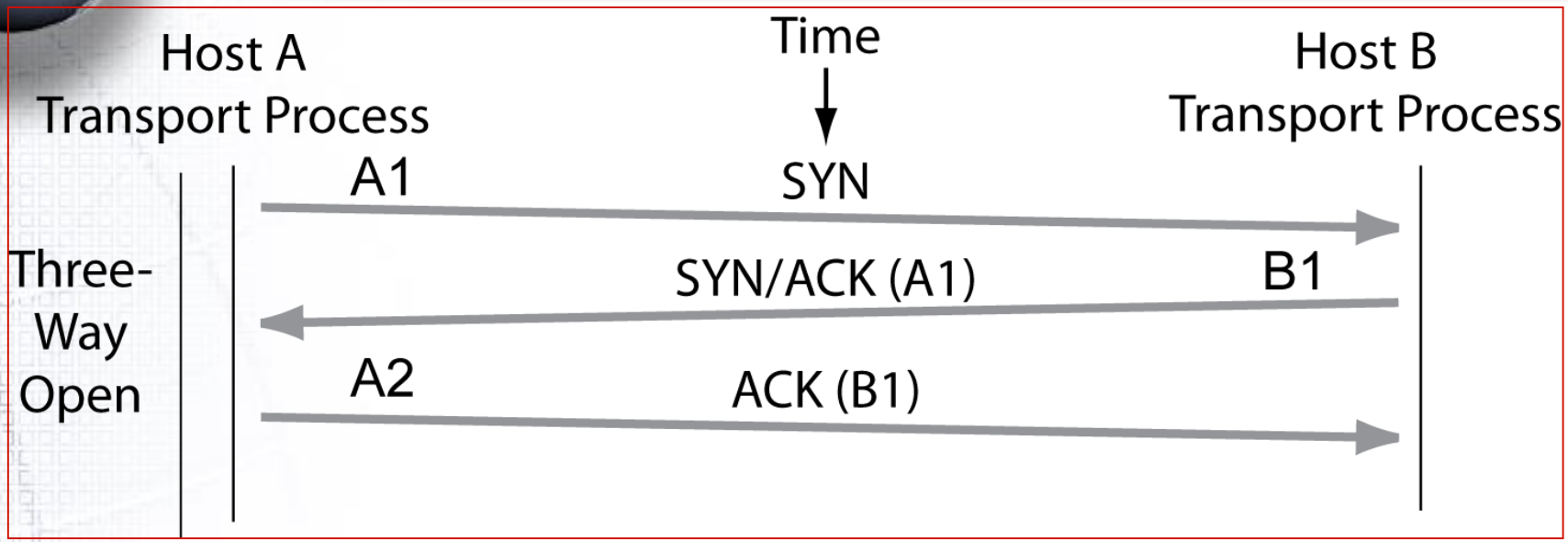
General Purpose (Core Later)	Layer	Specific Layer Purpose
Application-application communication	Application (5)	Application-application interworking
Transmission of a packet across an internet	Transport (4)	Host-host (process-to process) communication
	Internet (3)	Packet delivery across an internet
Transmission of a frame across a single network (LAN or WAN)	Data Link (2)	Frame delivery across a network
	Physical (1)	Device-device connection

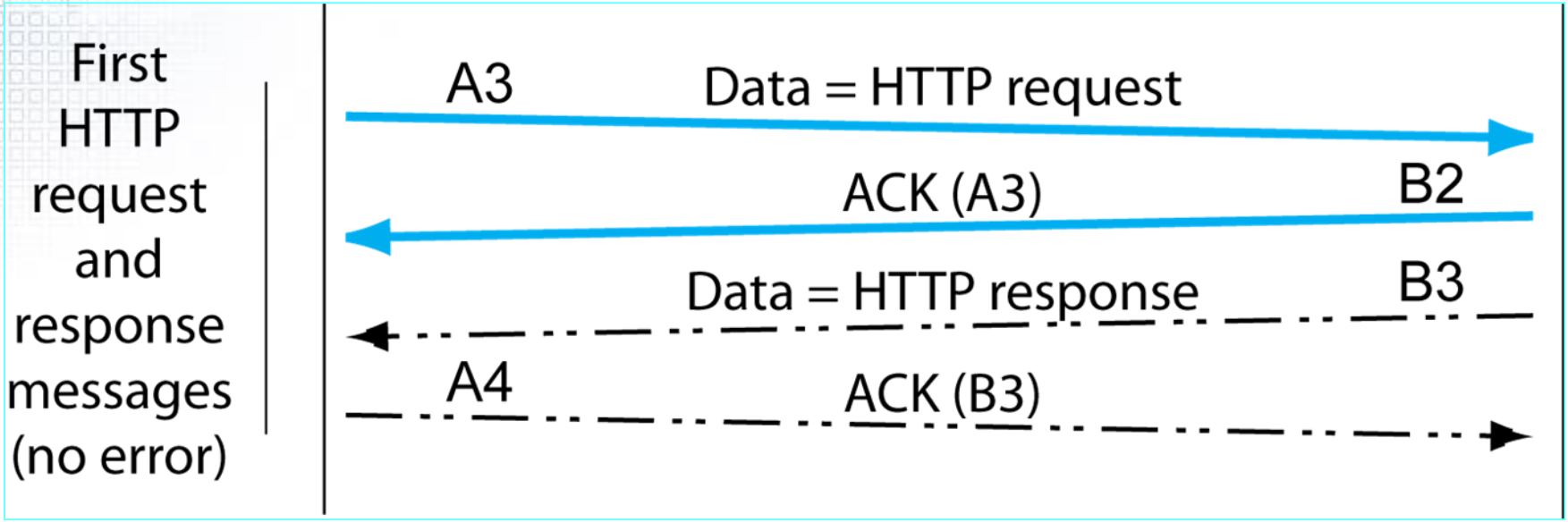
OSI

Network Standards

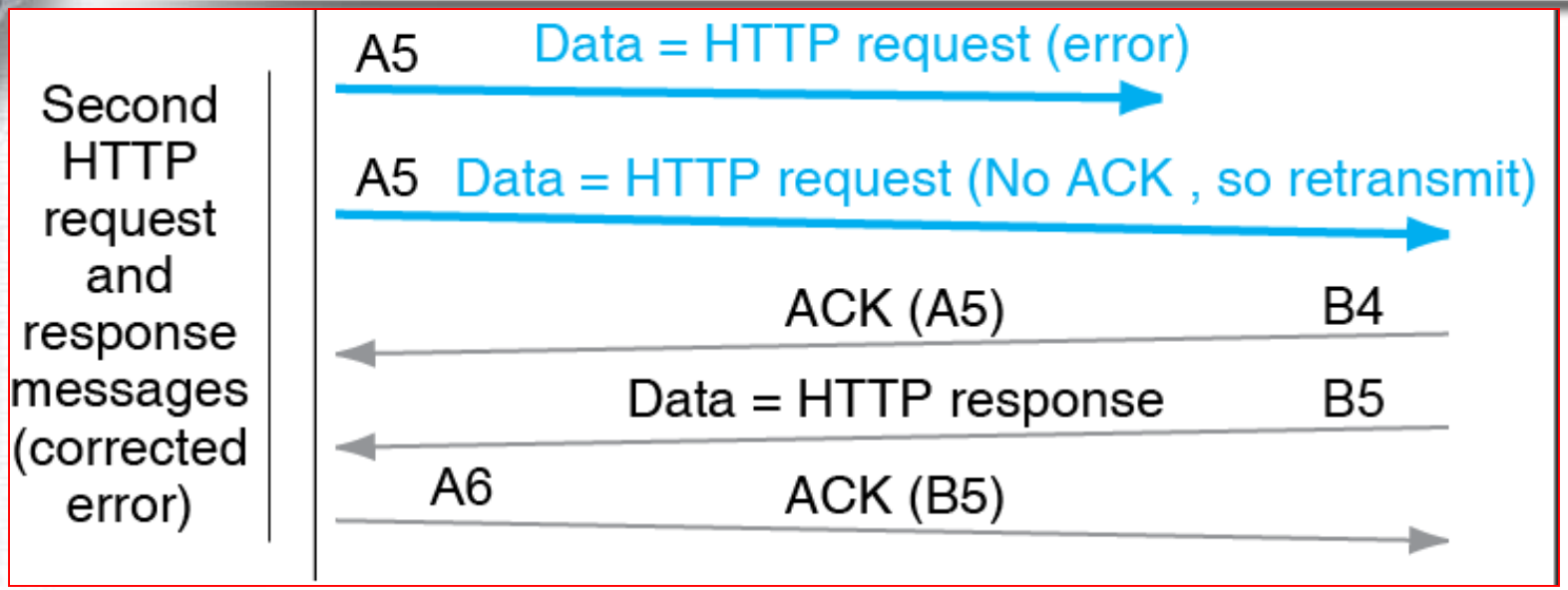
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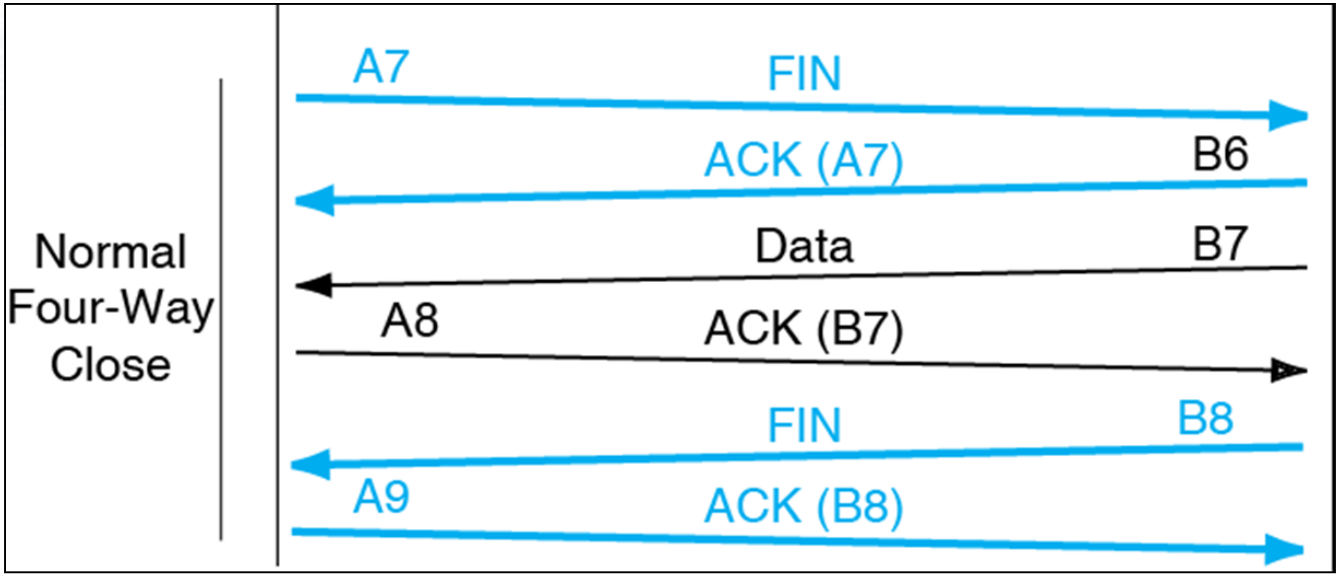
2.8 A TCP Session (1/2)





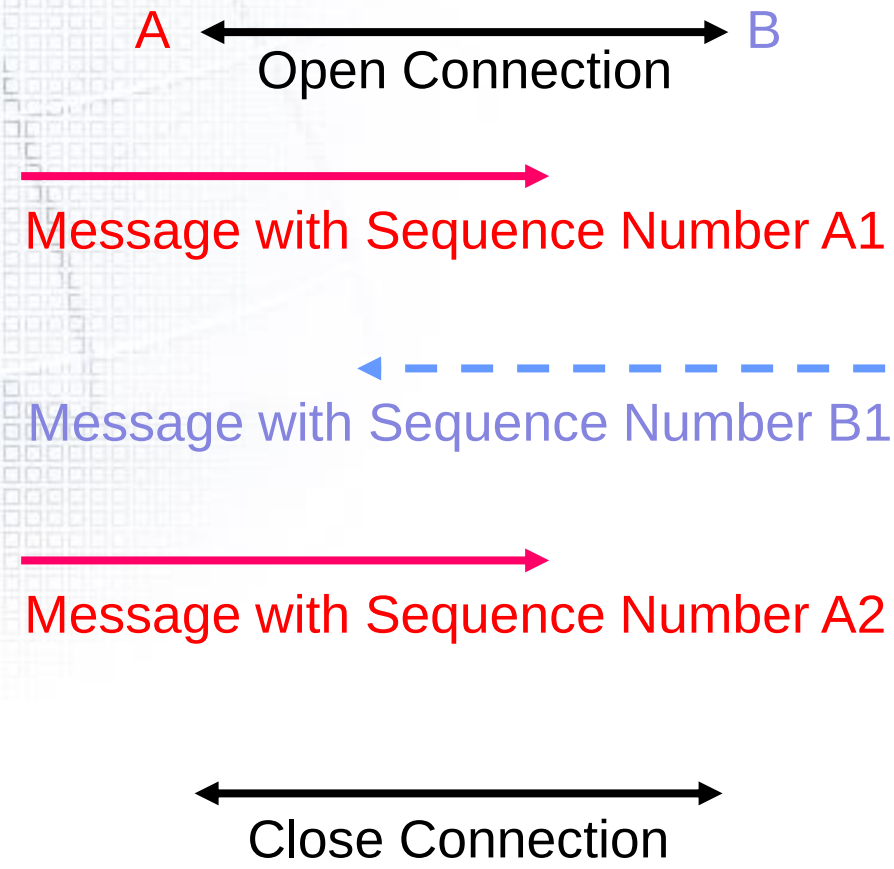
2.8 A TCP Session (2/2)



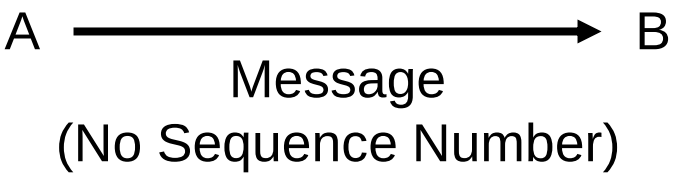


Important Concept in Message Ordering:
Connection-Oriented vs. Connectionless Protocols

Connection-Oriented Protocol



Connectionless Protocol



TCP	HTTP
Connection-Oriented	Connectionless
Openings, closings, ACKs place a **heavy load** on the network	No openings, closings, ACKs Creates a light load on the network
Reliable	Unreliable

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General Message Syntax (Organization)

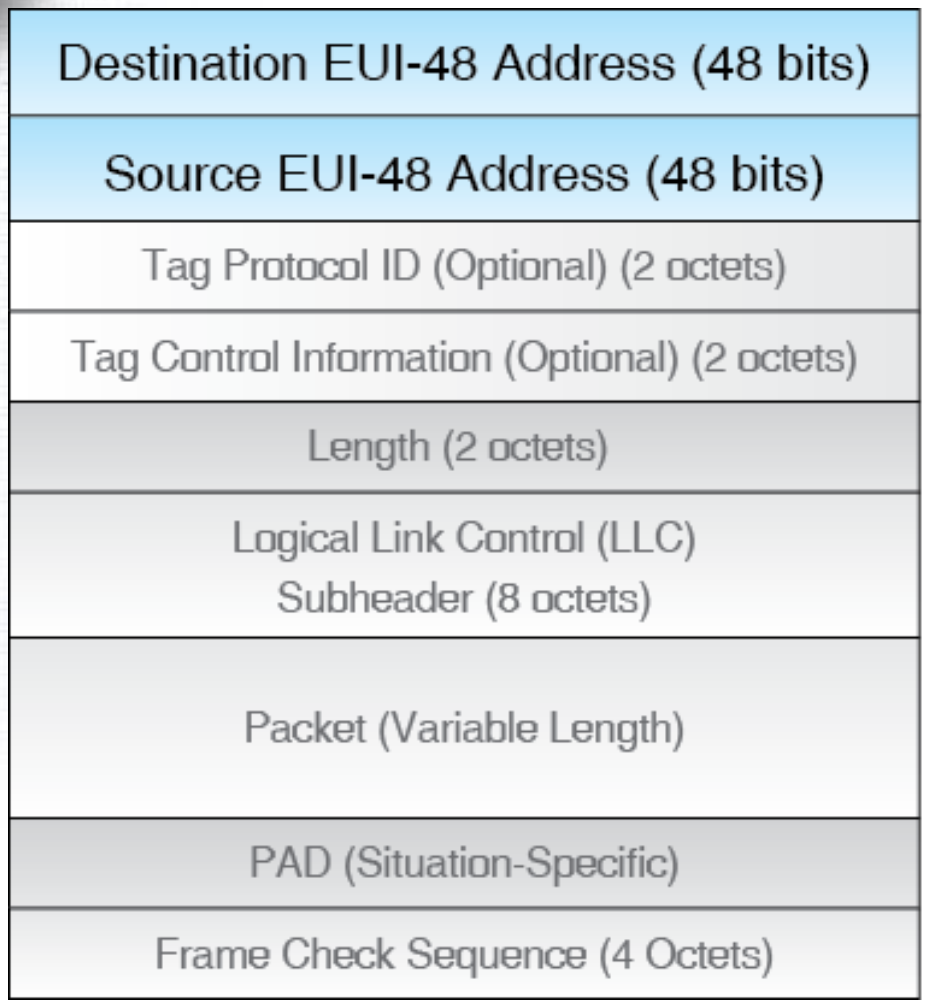
- General Message Organization
- Primary parts of messages
  - Data Field (content to be delivered)
  - Header (everything before the data field)
  - Trailer (everything after the data field)
- The header and trailer act like a delivery envelope for the data field



Q:是否每一個protocol message都必須由此三部分組成？

Answer

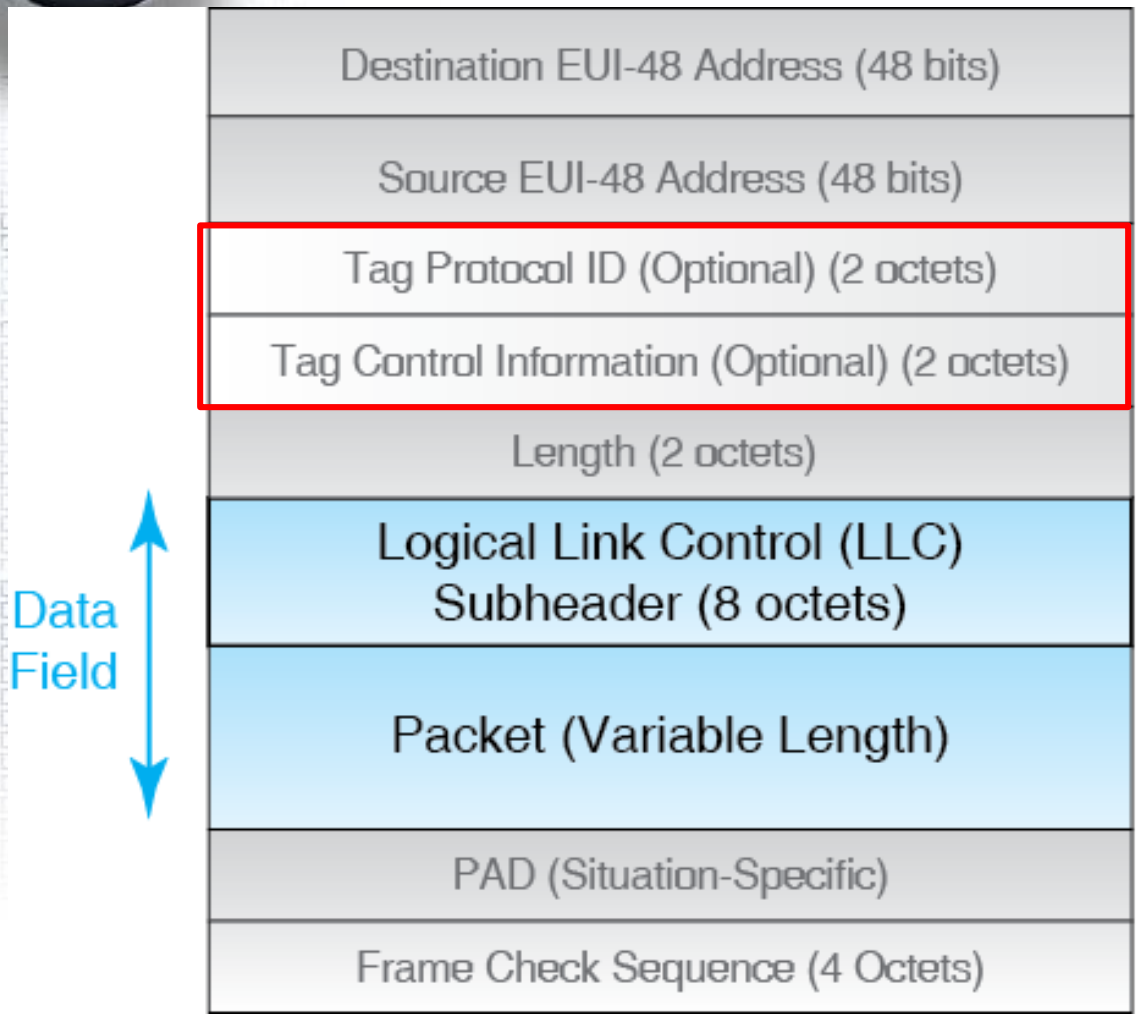
2.10 Ethernet Frame (1/3)

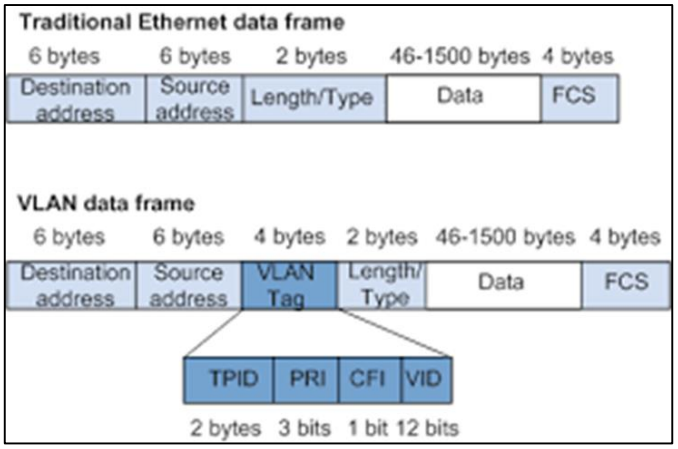


Destination and Source Data Link Layer Addresses (EUI-48). Formerly called a MAC Address

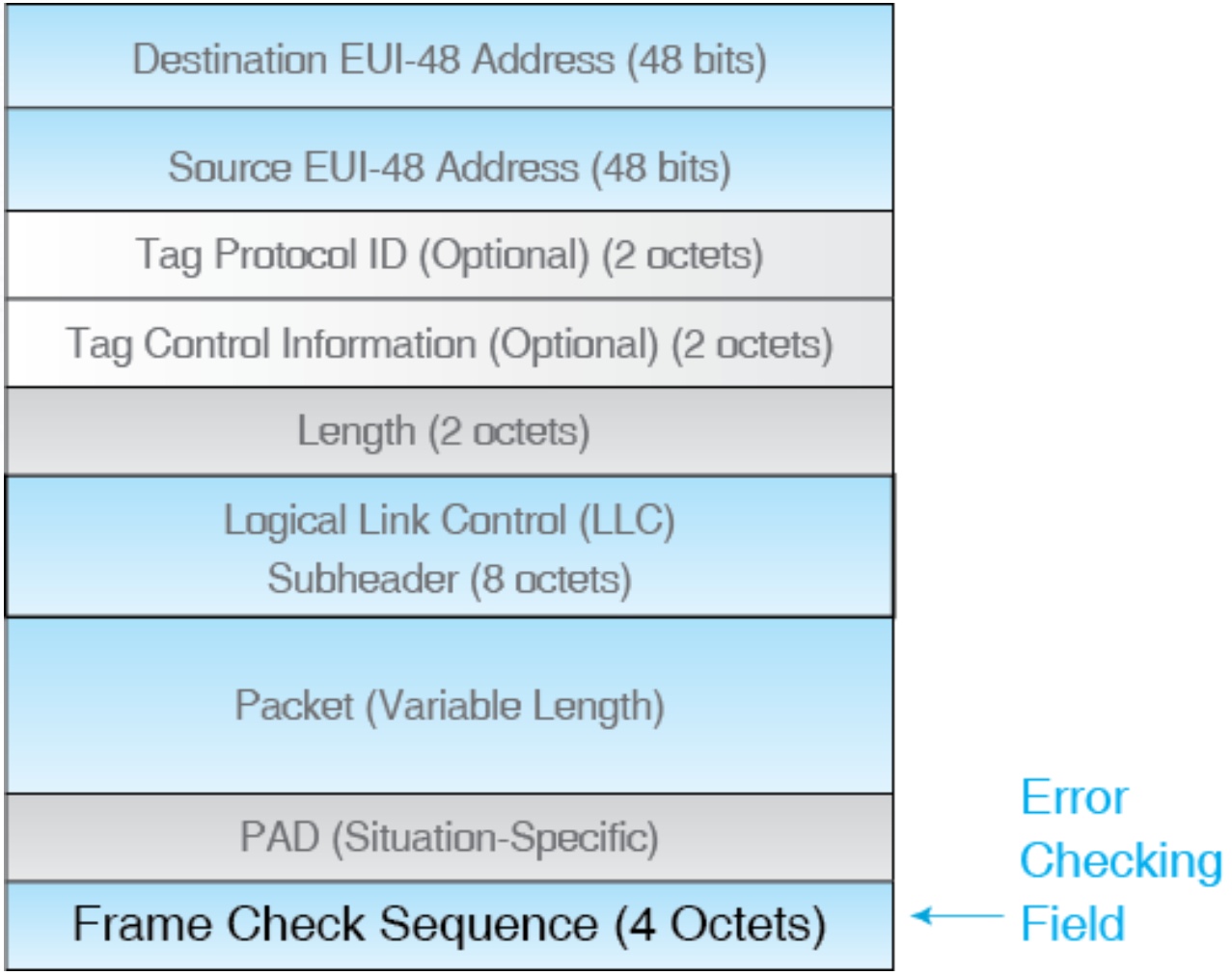
In Ethernet frame syntax, **fields are listed in order**, each is a certain number of octets (bytes) long

2.10 Ethernet Frame (2/3)

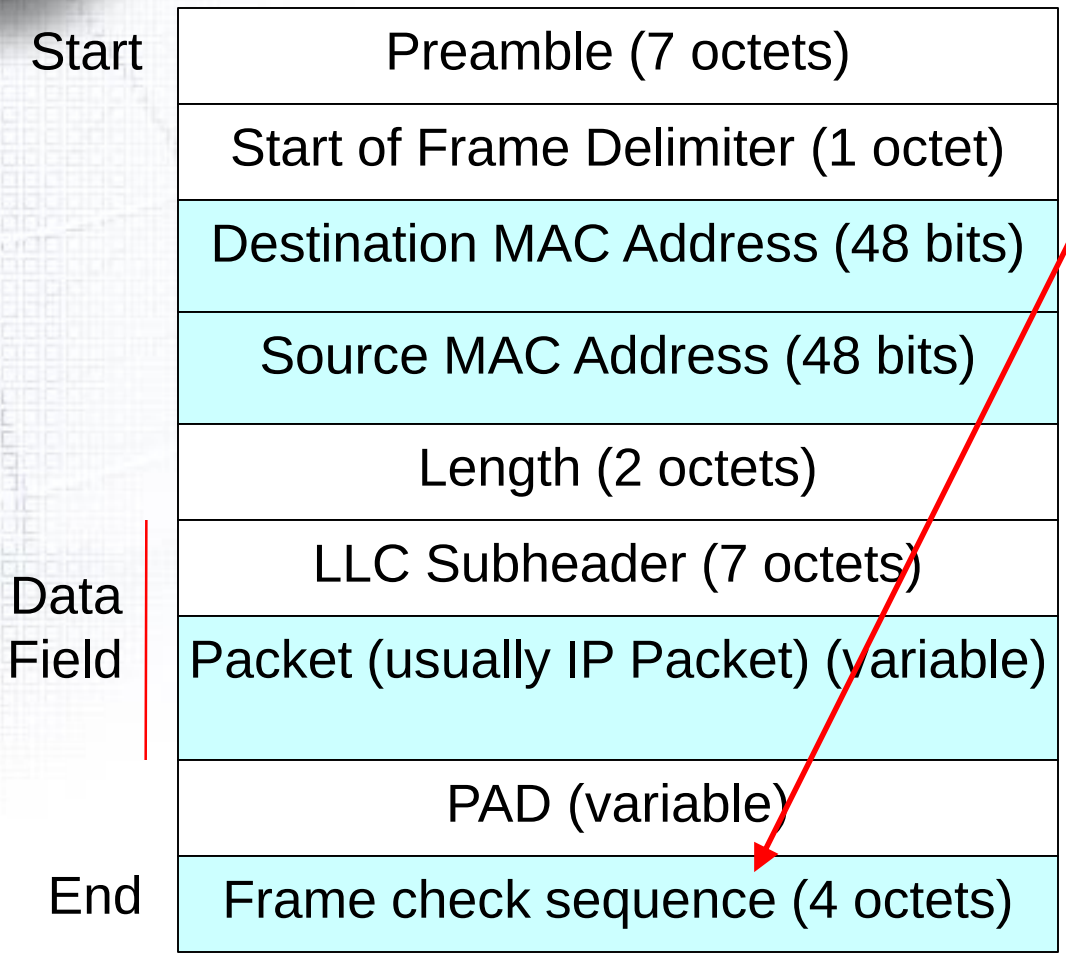




2.10 Ethernet Frame (3/3)



Ethernet Frame Summary



Octet = 8 Bits
10010111

Receiver uses Frame check sequence field to check for transmission errors

If an error is detected, the receiver merely discards the frame

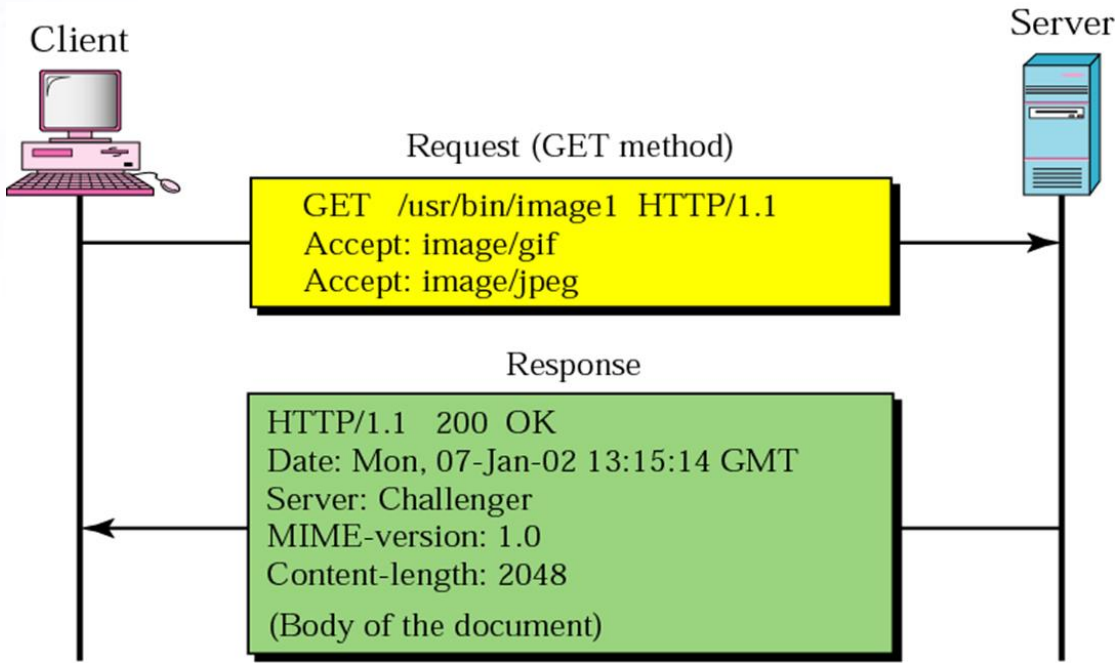
This is error detection

No retransmission, so no error correction

Ethernet is not reliable

2.15 HTTP Message Syntax

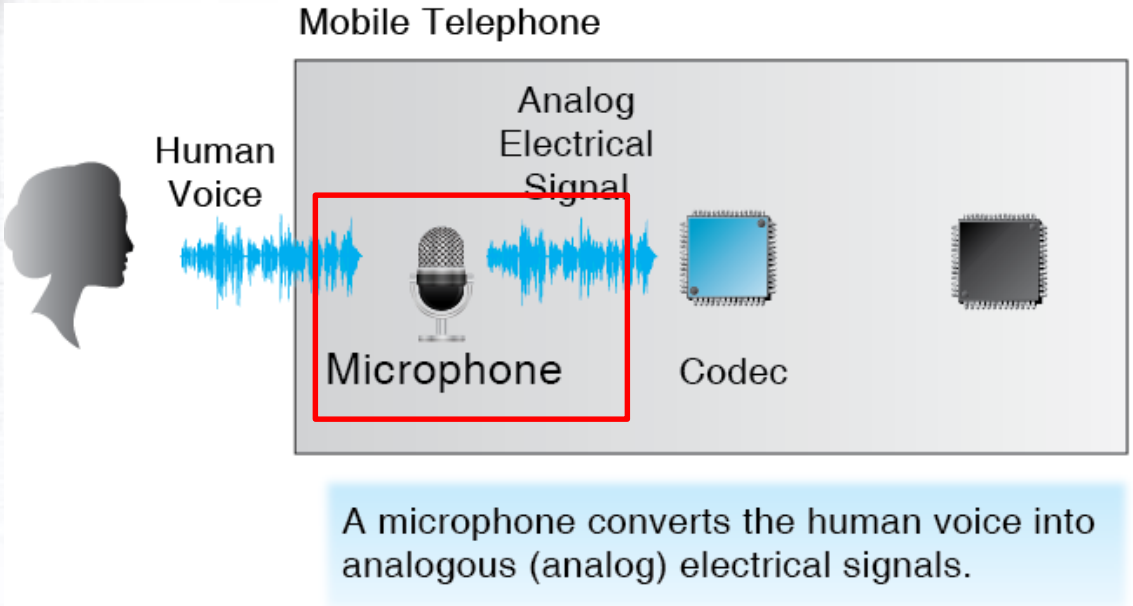
- HTTP is a relatively **old feeling** protocol
- Fields ended by **CRLF**, which starts a new line
- Based on e-mail (an old protocol) for **rapid** development
- Many application layer protocols, such as SMTP for e-mail, are much **more complex than HTTP**



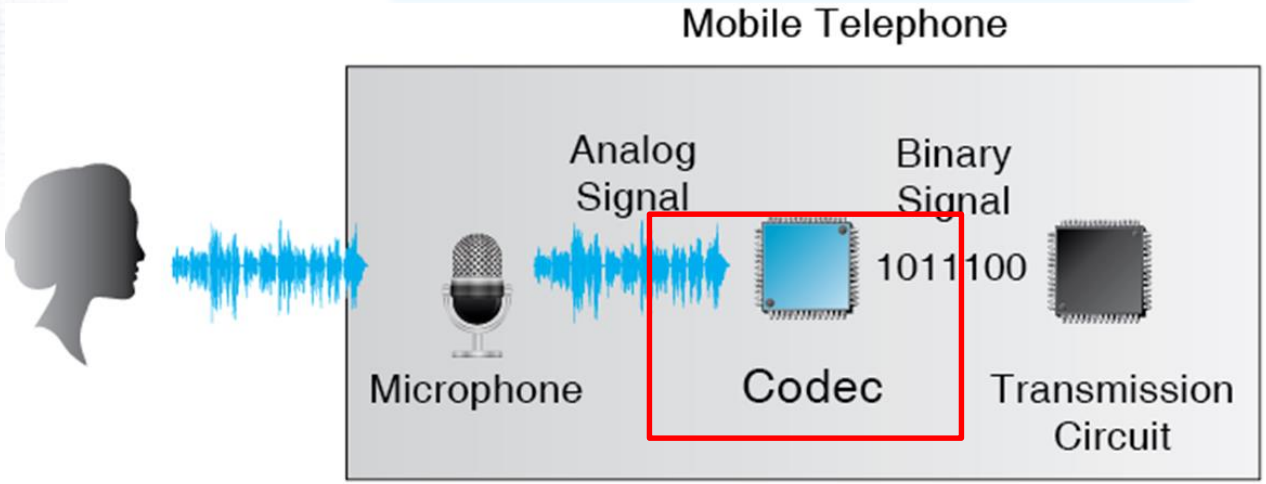


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2.20 Encoding Voice



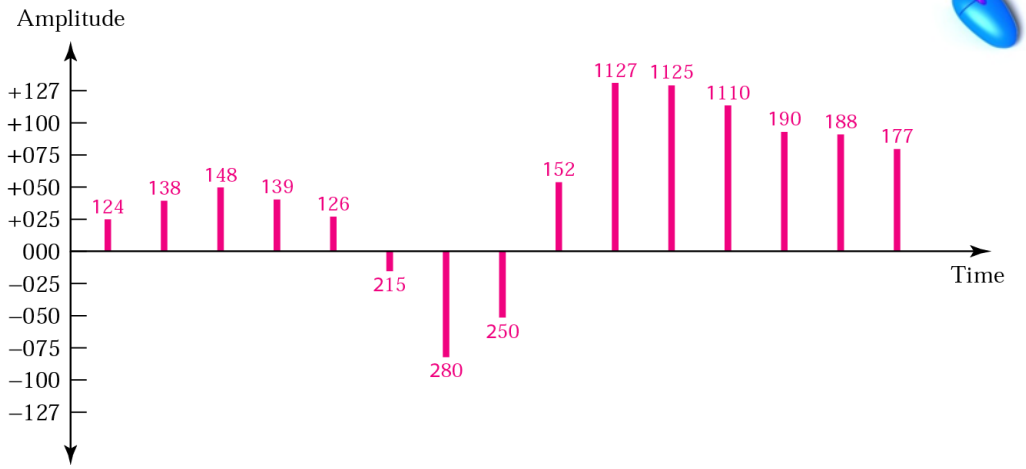
A microphone converts the human voice into analogous (analog) electrical signals.



The codec converts between analog electrical signals and digital transmission signals.

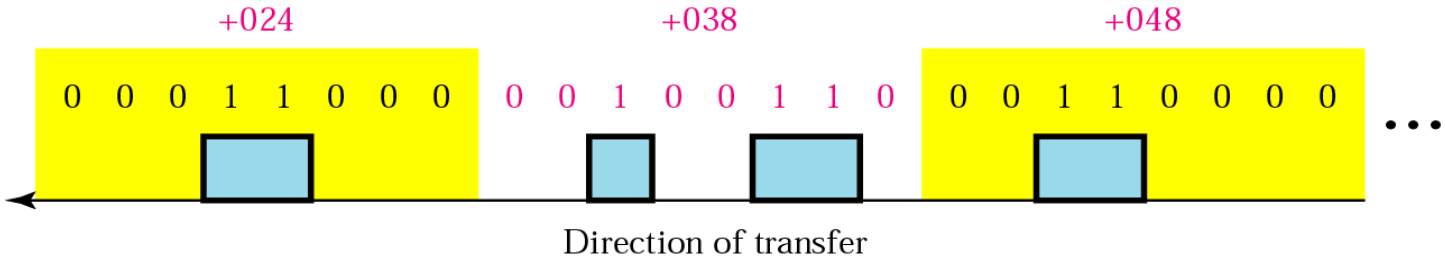
複習：PCM (Pulse Code Modulation)

- **Steps**
  - **Sampling**
  - **Quantization**
  - **Binary encoding**
  - **Line coding**



+024	00011000	–015	10001111	+125	01111101
+038	00100110	–080	11010000	+110	01101110
+048	00110000	–050	10110010	+090	01011010
+039	00100111	+052	00110110	+088	01011000
+026	00011010	+127	01111111	+077	01001101

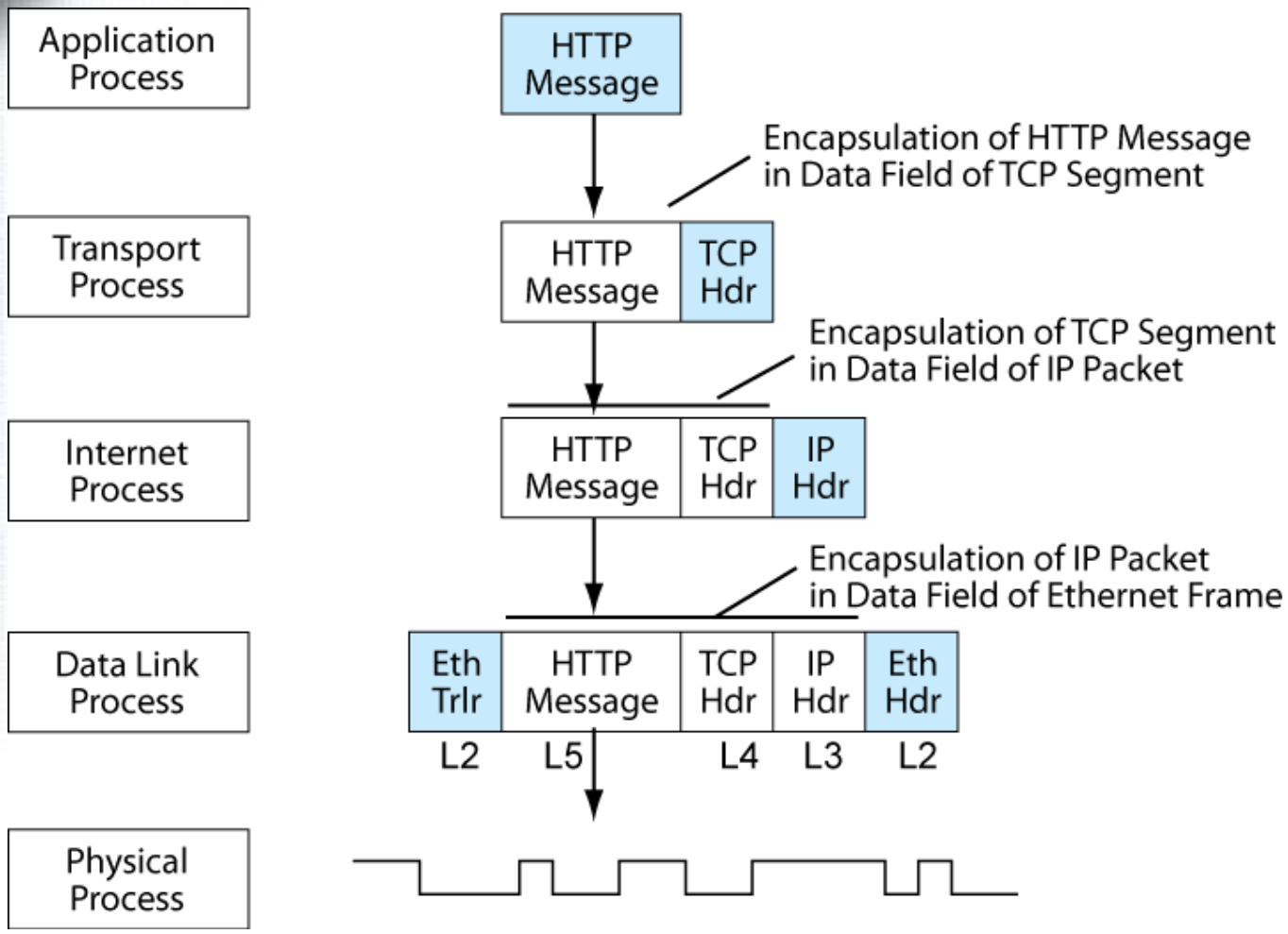
Sign bit
+ is 0 – is 1





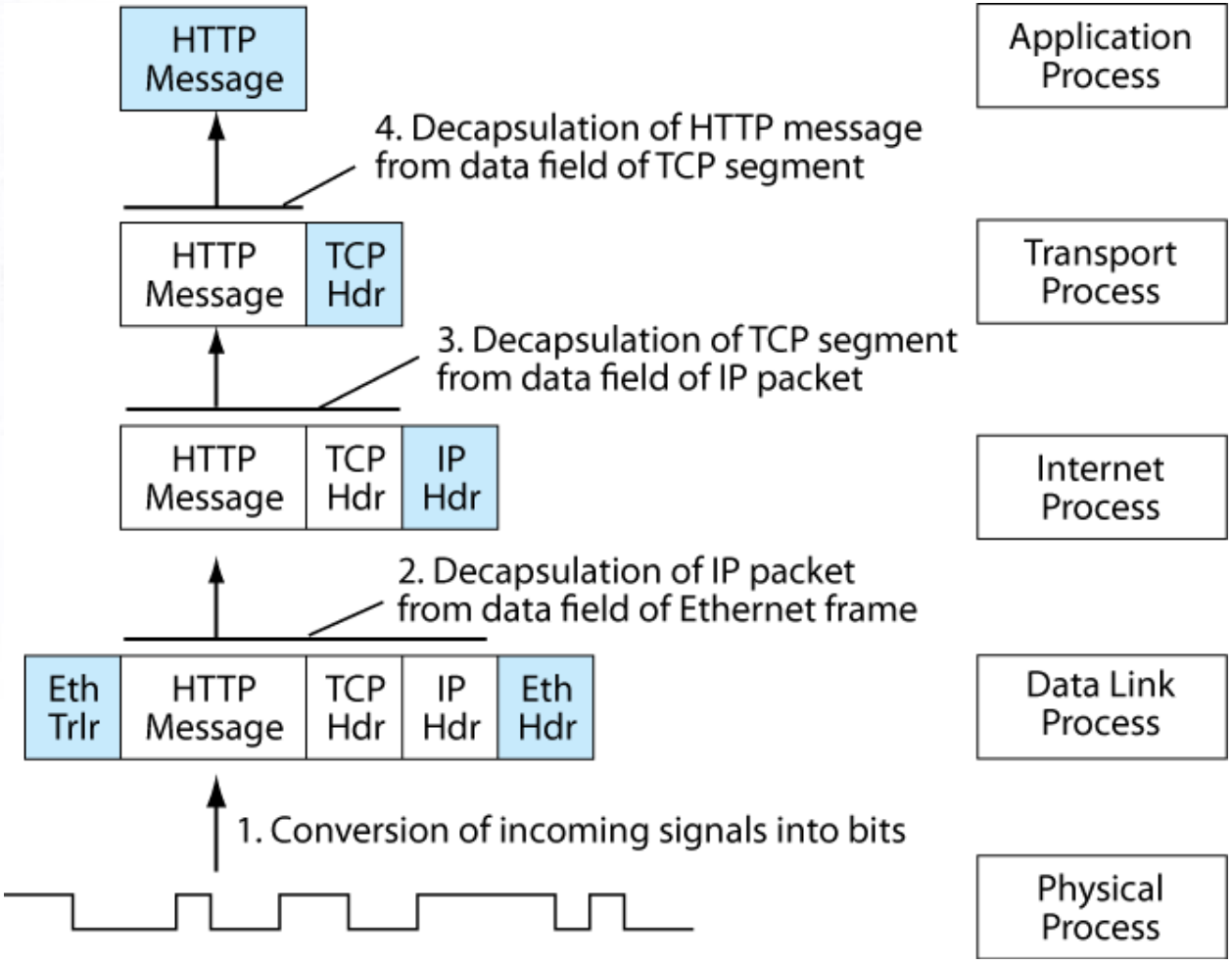
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Layered Communication on the Source Host

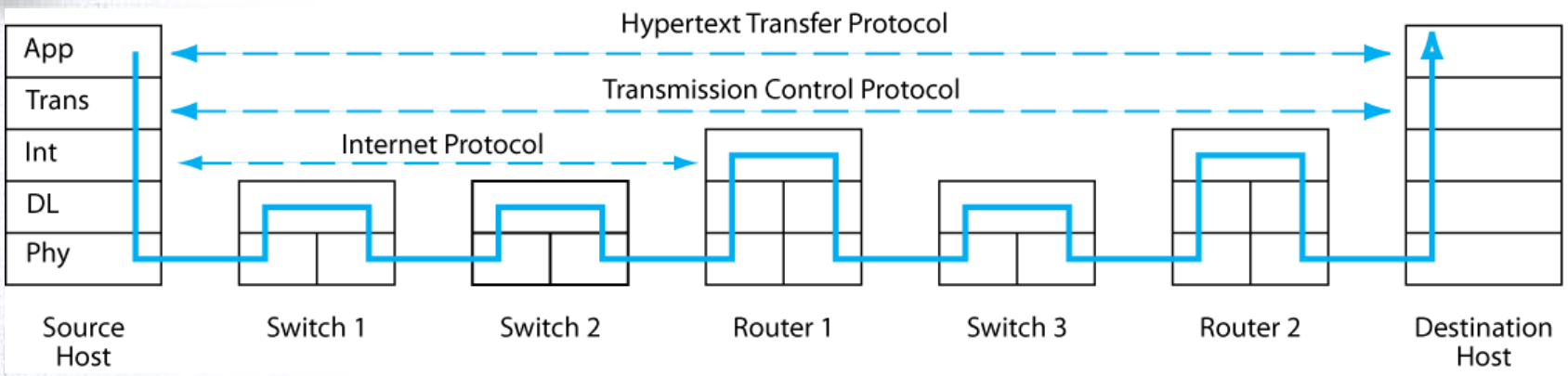


Each laye
a process
or software)

Decapsulation on the Destination Host



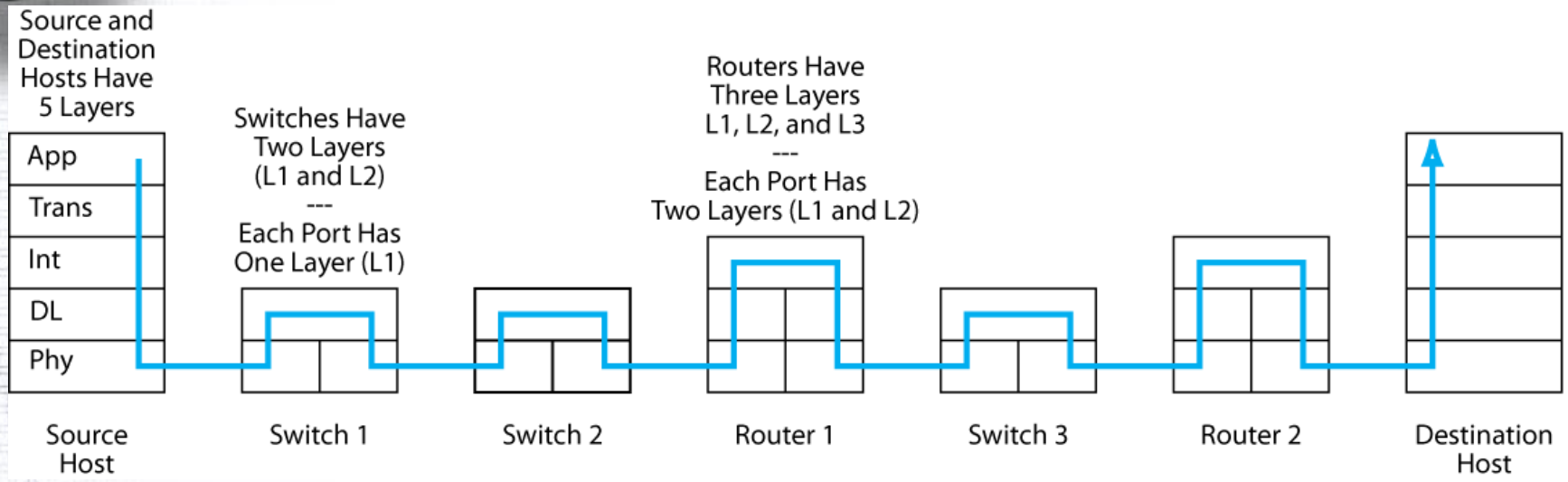
Vertical and Horizontal Communication



1. Not All Devices Have All Layers
    1. Hosts have all five
    2. Routers have only the lowest three

    Switches have only the lowest two

2-17: Layered End-to-End Communication



Encapsulation and decapsulation also occurs on each switch and router along the way

In switches, the highest layer is the data link layer, So switches are called Layer 2 devices

On routers, the highest layer is the internet layer, So routers are called Layer 3 devices

2.22 Protocols in This Chapter

Layer	Protocol	Connection-Oriented or Connectionless?	Reliable or Unreliable?
5 (Application)	HTTP	Connectionless	Unreliable
4 (Transport)	TCP	Connection-oriented	Reliable
4 (Transport)	UDP	Connectionless	Unreliable
3 (Internet)	IP	Connectionless	Unreliable
2 (Data Link)	Ethernet	Connectionless	Unreliable

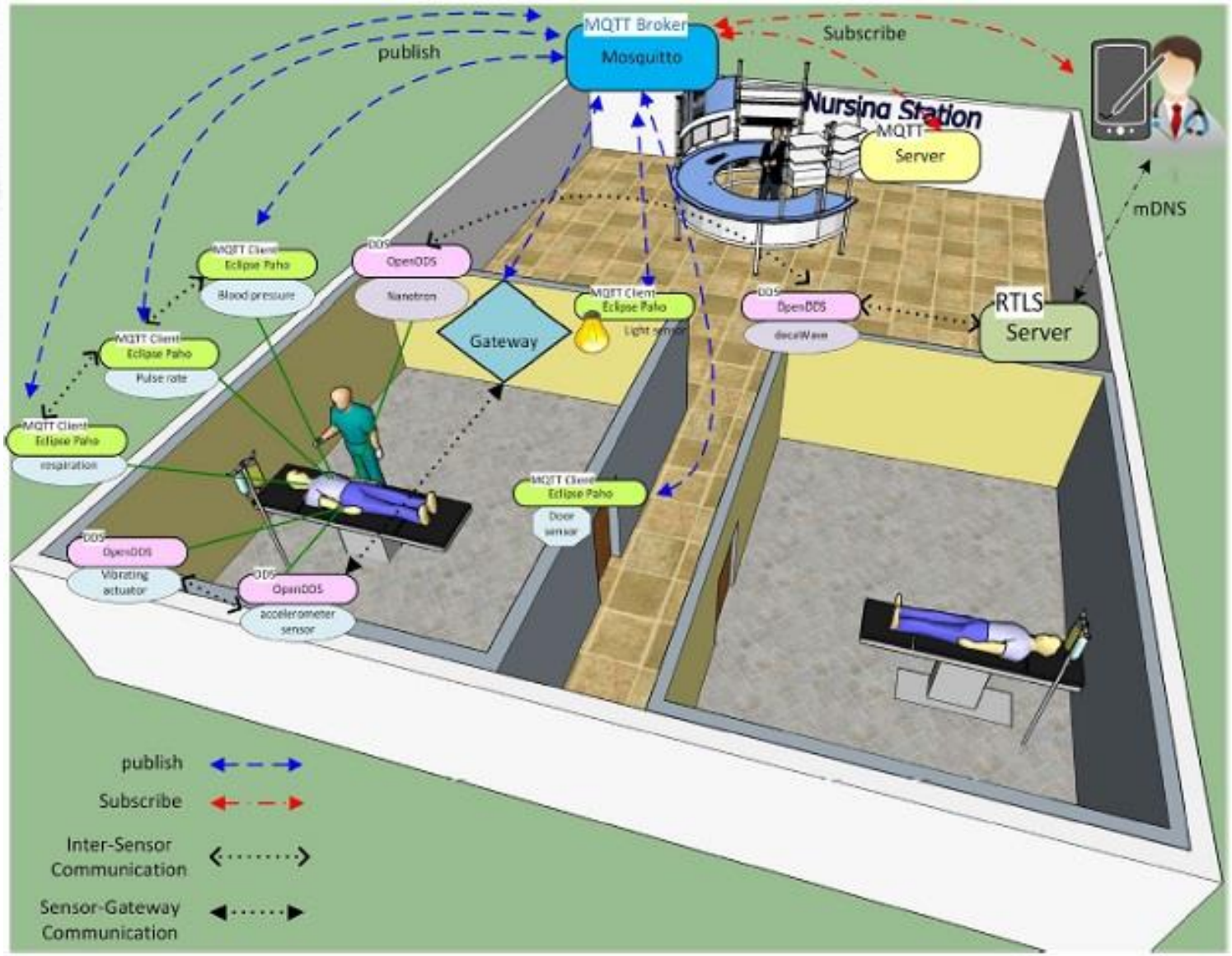
Summary of Network Standards

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以下為補充教材，取材自

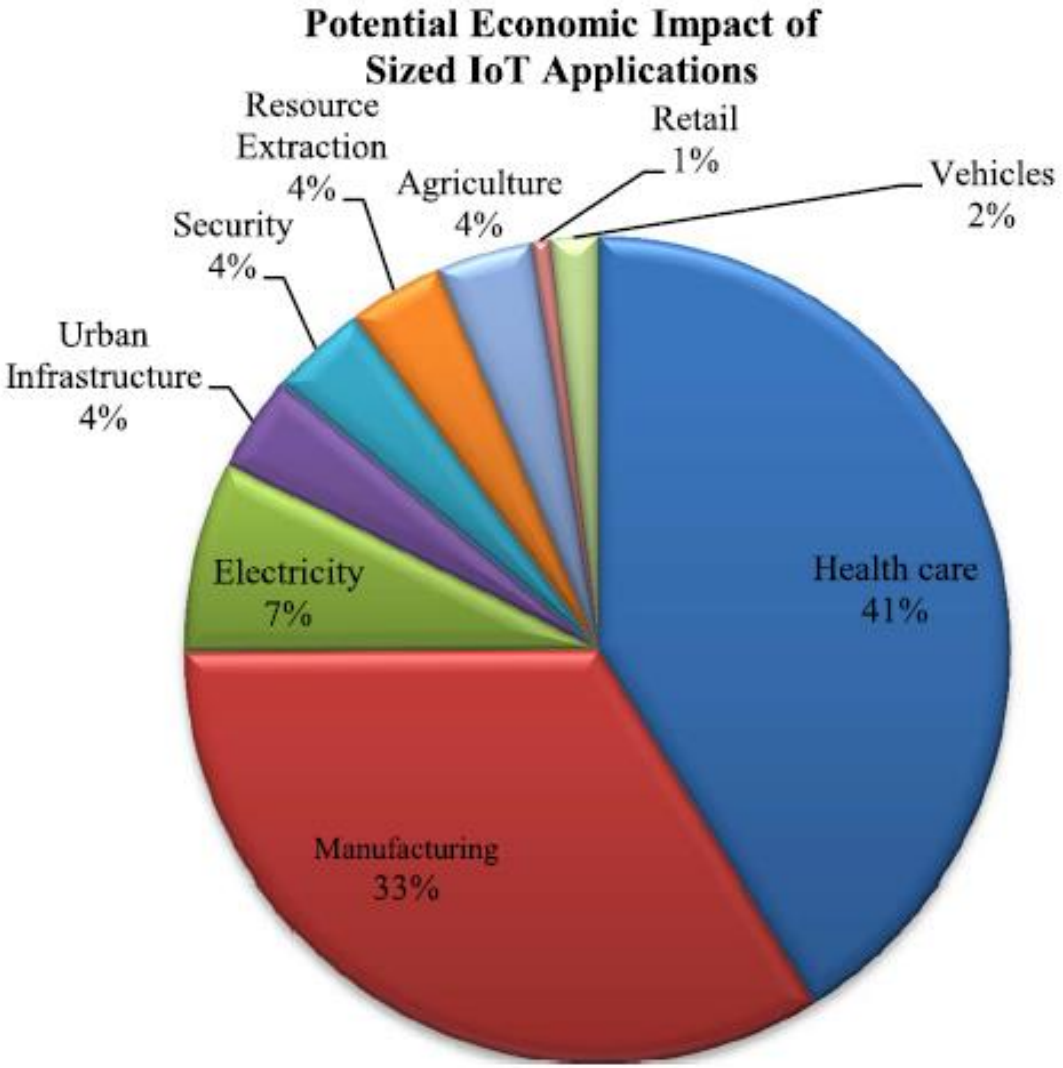
Ala Al-Fuqaha, et al.,” **Internet of Things: A Survey on Enabling Technologies**, **Protocols, and Applications**”, *IEEE COMMUNICATION SURVEYS & TUTORIALS*, VOL. 17, NO. 4, 2015, pp.2347~2376

Application Example of Internet of Things (IoT)



Market Share of IoT Domains

-



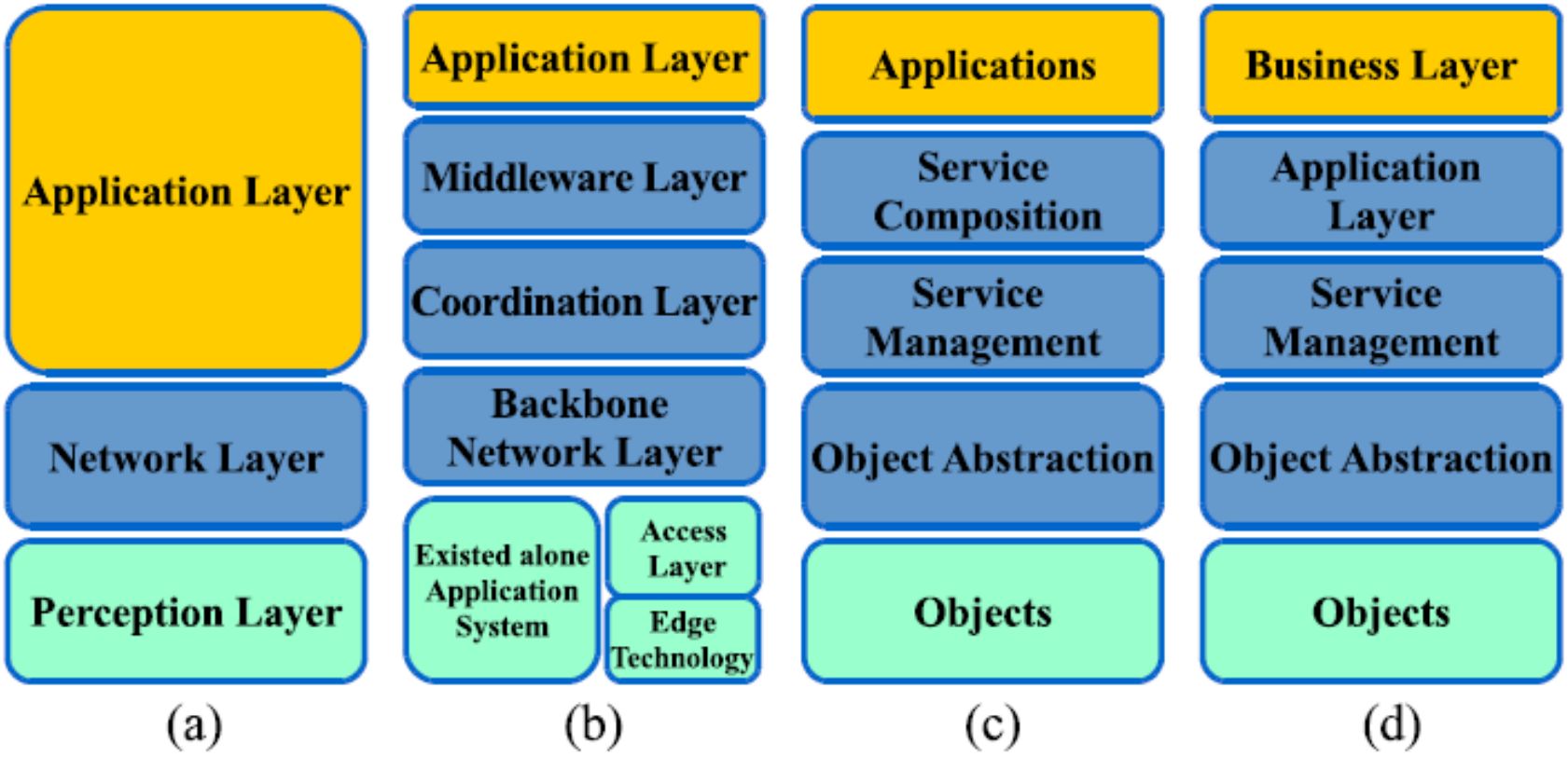
IoT Elements



Services:

1. Identity-related Service
2. Information-Aggregation Service
3. Collaborative-Aware Service
4. Ubiquitous Service (Anytime, Anywhere)

Various Types of IoT Architecture



Common Protocols for IoT

TABLE III
STANDARDIZATION EFFORTS IN SUPPORT OF THE IOT

Application Protocol		DDS	CoAP	AMQP	MQTT	MQTT-SN	XMPP	HTTP REST
Service Discovery		mDNS				DNS-SD		
Infrastructure Protocols	Routing Protocol	RPL						
	Network Layer	6LoWPAN				IPv4/IPv6		
	Link Layer	IEEE 802.15.4						
	Physical/ Device Layer	LTE-A	EPCglobal	IEEE 802.15.4		Z-Wave		
Influential Protocols		IEEE 1888.3, IPSec				IEEE 1905.1		

MQTT (Message Queue Telemetry Transport) Architecture and AMQP (Advanced Message Queueing Protocol) Frame

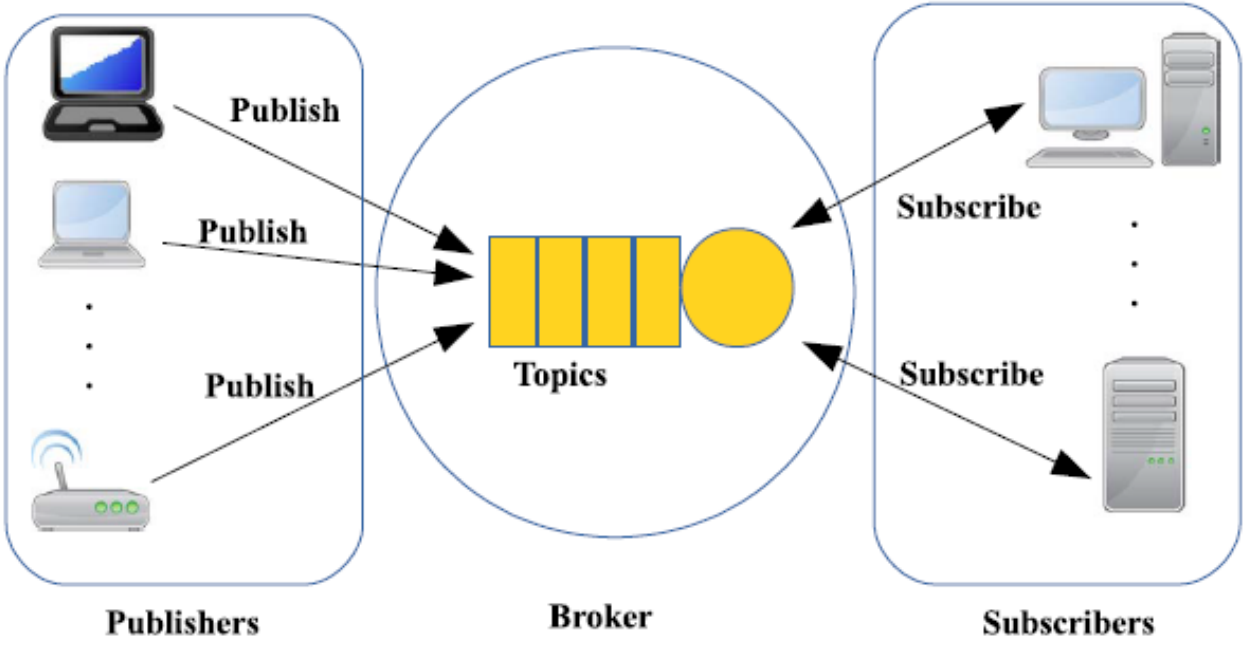


Fig. 8. The architecture of MQTT.

DOFF: data offset

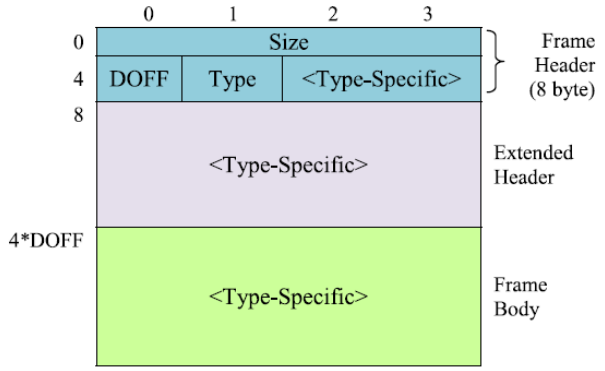


Fig. 15. AMQP frame format.

mDNS (multicast DNS) and DNS-SD (DNS Service Discovery)

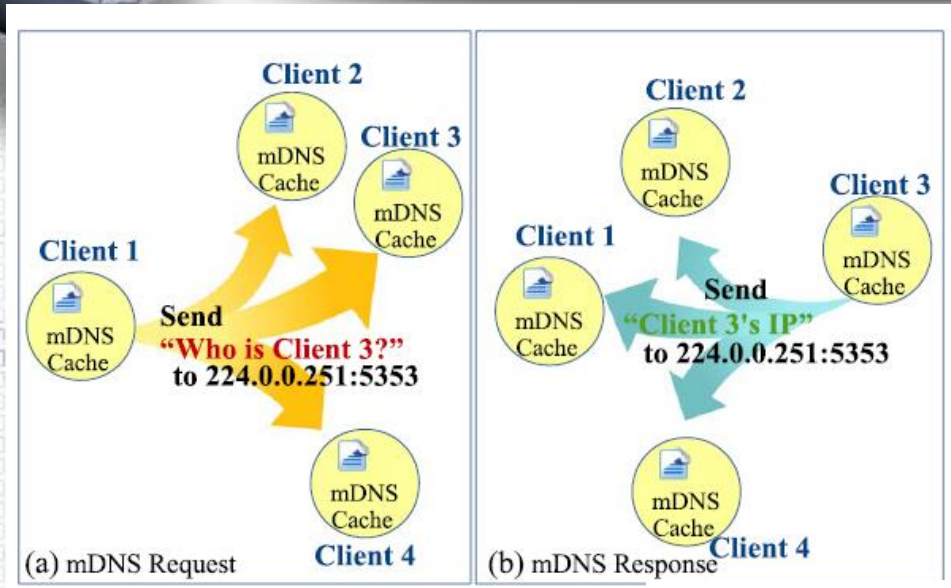


Fig. 17. Request/Response in mDNS protocol.

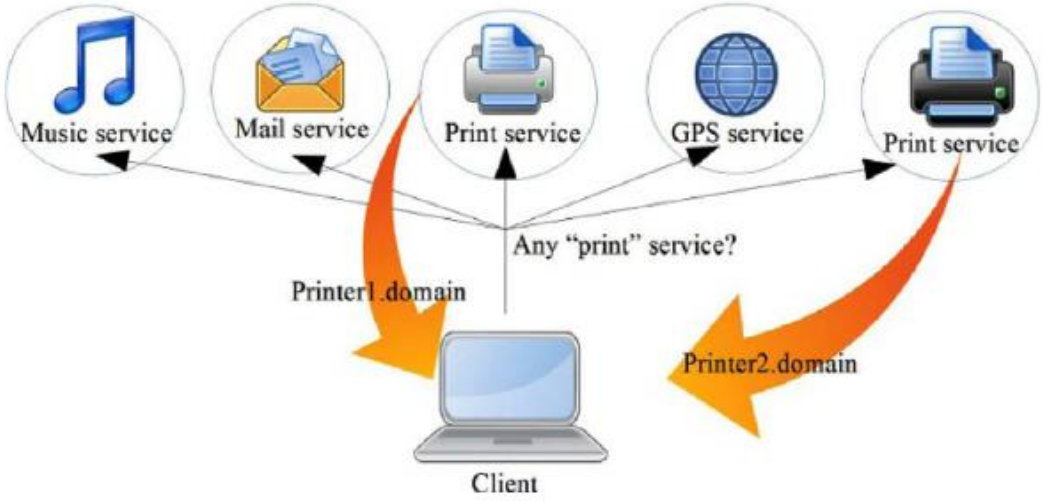


Fig. 18. Discovering print service by DNS-SD.

第二章分組題目

一. 請閱讀下列兩篇文章並作綜合報告以回答下列問題

1) “**Cross Layer Design: A Survey and the Road Ahead**”, by V. Srivastava, et al, in IEEE Communications Magazine, December 2005

2) “**Context Awareness Through Cross-Layer Network Architecture**”, by M. Razzaque, et al, 2007.

Q:請綜合說明此兩篇文章的重點後，先回答非分層架構 (Non Layered Structure) 適用於何種情況？然後詳述此跨層架構設計的概念對現代AI系統的設計帶來何種影響？